

MODEL QUESTION PAPER

TED(15)3051 THIRD SEMESTER DIPLOMA EXAMINATION IN AUTOMOBILE ENGINEERING

Revision 2015

FLUID MECHANICS AND MACHINERY

Max.marks 100

Time 3 Hrs.

PART A

(Maximum Marks 10)

I Answer the following questions in one or two sentences.

1. Define kinematic viscosity.
2. State Newton's equation for viscosity.
3. Define slip of a reciprocating pump.
- 4 Name any two pneumatic actuators.
5. Define flash point.

(2 x5=10)

PART B

(Max. marks 30)

II (Answer any five of the following questions. Each question carries 6marks)

1. List any six properties of fluids.
2. Explain the different methods of pressure measurement.
3. List and explain different types of losses in flow through pipes.
4. Explain the classifications of orifices.
5. Explain the working of centrifugal pump with a sketch.
6. Compare pneumatic system and hydraulic system.
7. Explain the working of a poppet valve.

(6 x 5= 30)

PART C

(Max.marks 60)

(Answer any one full question from each unit)

UNIT I

III (a) A differential manometer is used to connect two pipes A and B. The pipe A contains a liquid of Sp. Gravity 1.1 under a pressure of 150 kN/m^2 . The left limb of the manometer is connected to pipe A which is 1.6 meter above the pipe B. The pipe B contains oil of Sp. gravity 0.8 under a pressure of 250 kN/m^2 . The level of mercury in the left limb is 1.5m below level of pipe B the level of mercury in the right limb is below to that in the left limb. Find the difference between the mercury levels. (8)

b. Differentiate between simple and differential manometers. (7)

OR

IV (a) Distinguish the terms atmospheric pressure, gauge pressure, vacuum pressure and absolute pressure. (8)

(b) If 5 m^3 of oil weighs kN, calculate its specific weight, specific volume, mass density and relative density (7)

UNIT II

V(a) A horizontal venturimeter is provided in a pipeline 300mm diameter carrying oil of Sp. Gravity 0.8. The throat diameter is 150mm. If the differential manometer gives a reading of 250mm of mercury find the discharge in liters per second. Assume coefficient of discharge as 0.97 (8)

(b) State and explain Bernoulli's theorem. (7)

OR

VI (a) Explain different types of fluid flow (8)

(b) A pipe 500m long pipe is conveying water with a velocity of 10 m/s. Find the

Suitable diameter of the pipe if the head loss due to friction is 3.4m. Take $f = 0.001$ (7)

UNIT III

VII (a) A double acting reciprocating pump has a stroke 300 mm, piston diameter 150mm, delivery head 20m and suction head 3m. If the pump is working at 60 rpm the actual and theoretical power required to drive the pump, assume the efficiency as 75%. (8)

(b) Explain with sketch the working of multistage pumps. (7)

OR

VII(a) Explain the working of a lobe pump with a neat sketch (8)

(b) Explain different efficiencies of a pump. (7)

UNIT IV

IX(a) Explain the basic components of a hydraulic system with a neat diagram (8)

(b) Describe the working of an air filter in air line. (7)

OR

X(a) Explain the operation of a manually operated three way valve (8)

(b) Explain the working of a solenoid operated valve. (7)

